

Supporting Information

Rapid sequential touches complicate whisking-phase analysis in mouse active touch

Yuxuan Xu

Technical details

Extraction and timing

All event-level variables were reconstructed from the versioned source files in DANDI:000013, version 0.220126.2143. The source table contained 41,428 touch events from 52 files. Intracellular-electrode metadata identified 41 layer 4 files and 11 layer 5 files. Applying the layer, trial-time, stimulation, and phase criteria yielded 19,675 touches from 21 archive subject labels. The complete archive contains 23 labels. The source article reports 21 mice across 52 recording sessions, and no crosswalk verifies a one-to-one mapping. Sex is recorded as unknown in the NWB subject metadata.

Spike counts were reconstructed from the 10-kHz spike trains with half-open intervals. Previous and following onset intervals were recomputed within trials and rounded to 0.001 ms before thresholding. At a trial boundary, an absent preceding or following touch was treated as no neighboring onset. The response-window analysis excluded a touch when the following onset fell inside the current response window. Consequently, the 10-ms comparison selected 16,950 touches and modeled 16,412 touches from 38 files. The 50-ms comparison selected 13,692 touches and modeled 13,360 touches from 36 files.

Whisking amplitude was calculated as the Hilbert-envelope magnitude of the archived filtered base-angle signal within contiguous video segments. Velocity and acceleration were evaluated 5 ms before contact. They were estimated from the archived base angle with an 11-sample, third-order Savitzky-Golay filter and first or second derivatives in seconds. Peak curvature change was the largest signed or absolute change in the archived curvature signal during 0-20 ms after touch. Contact duration used paired onset and offset markers.

Model implementation

The context predictors contained $\log(1 + \text{touch number})$, first-touch status, log preceding interval after values below 1 ms were set to 1 ms, pole position, time from pole entry, and time within the recording. The measured-kinematic predictors added $\log(1 + \text{whisking amplitude})$, $\text{sign}(x)\log(1 + |x|)$ pre-touch angular velocity and acceleration, distance to pole at touch, and base angle at touch. The primary phase terms were sine and cosine of phase and sine and cosine of twice the phase. The separate-branch model included a retraction indicator and linear and quadratic progression terms for negative protraction phases and nonnegative retraction phases. A descriptive sensitivity model added $\text{sign}(x)\log(1 + 1,000|x|)$ signed curvature change, $\log(1 + 1,000|x|)$ absolute curvature change, and $\log(1 + \text{duration in milliseconds})$.

Trial-grouped cross-validation used up to five deterministic folds. Median imputation and predictor standardization were fit on each training fold. The Poisson model used an L2 penalty with $\alpha = 1$ for the primary analysis; the primary two-harmonic model was also evaluated at 0.01, 0.1, and 10. The logistic model used the corresponding inverse penalty parameter $C = 1/\alpha$. Null predictions were training-fold outcome means. The primary 50-ms phase set contained 12,773 selected touches. The phase coverage rules retained 12,296 touches from 35 files and 20 archive subject labels for model comparison.

The continuous-coefficient analysis used ordinary least squares after standardizing the outcome and all predictors within each file. The model included the context variables, two phase harmonics, and all eight movement or contact variables. Complete observations comprised 10,370 touches from 34 files and 20 archive subject labels. File coefficients were reduced to one median for each archive subject label.

Additional results

Supplemental Table S1. Isolation from both neighboring touch onsets

The same minimum interval was applied to the preceding and following touch onsets. Values summarize the mean of archive-label median file means for spike counts during 0-50 ms after touch. The 250-ms condition describes contacts with no observed onset within 250 ms; its interval is wide because the 1,014 touches are sparse within files.

Minimum interval to both neighbors (ms)	Touches	Files	Archive subject labels	Mean spikes per touch	Bootstrap 95% CI
0	19,675	41	21	0.417	0.279 to 0.570
20	19,160	41	21	0.416	0.279 to 0.569
30	18,157	41	21	0.415	0.278 to 0.562
50	12,773	41	21	0.416	0.285 to 0.559
100	6,251	41	21	0.432	0.293 to 0.585
250	1,014	41	21	0.514	0.291 to 0.791

Of the 1,014 touches in the 250-ms condition, 664 occurred at a trial boundary and lacked at least one observed within-trial neighbor. A stricter sensitivity analysis required both neighboring intervals to be observed and at least 250 ms. It retained 350 touches from 38 files and 21 archive subject labels; the mean was 0.350 spikes per touch (95% CI, 0.214 to 0.506).

Supplemental Table S2. Neighbor-threshold sensitivity of phase prediction

Minimum interval to both neighbors (ms)	Retained touches	Archive labels, context model	Median phase gain, context	Median phase gain, kinematic adjustment
0	19,675	21	0.0124	0.0096
20	19,160	21	0.0124	0.0082
30	18,157	21	0.0134	0.0105
50	12,773	20	0.0152	0.0084
100	6,251	17	0.0086	0.0083
250	1,014	0	not estimated	not estimated

At 250 ms, no recording file met the requirements of at least 80 observations and at least 15 observations in each phase branch. The descriptive response remains available in Supplemental Table S1.

At the 50-ms threshold, a stricter phase sensitivity required an observed neighbor on each side. It selected 10,097 touches; 9,247 touches from 30 files and 19 archive subject labels met the phase-model criteria. Median phase gains were 0.0203 after contextual adjustment (95% CI, 0.0046 to 0.0336; $P < 0.001$) and 0.0079 after measured-kinematic adjustment (95% CI, 0.0008 to 0.0212; $P = 0.002$).

Supplemental Table S3. Baseline and response-window sensitivity

The table crosses all response and baseline windows. Dense-sequence estimates retained all preceding intervals; the comparison condition required a preceding interval of at least 100 ms. Q values control the false discovery rate across these 50 sensitivity estimates. Complete numerical results, sample counts, and confidence intervals are included with the numerical source data.

Response (ms)	Baseline (ms)	Dense beta	Dense q	100-ms beta	100-ms q
10	10	0.037	0.035	0.014	0.844
10	20	0.077	0.010	0.010	0.751
10	30	0.085	0.009	0.012	0.751
10	40	0.084	0.009	0.012	0.751
10	50	0.080	0.009	0.015	0.603
20	10	0.088	0.010	0.053	0.350
20	20	0.098	0.009	0.053	0.228
20	30	0.110	0.009	0.030	0.291
20	40	0.120	0.009	0.020	0.279
20	50	0.117	0.009	0.046	0.200
30	10	0.097	0.011	0.027	0.751
30	20	0.108	0.010	0.011	0.751
30	30	0.125	0.009	0.008	0.822
30	40	0.121	0.009	0.002	0.797
30	50	0.116	0.010	0.016	0.751
40	10	0.079	0.016	-0.006	0.813
40	20	0.116	0.015	-0.017	0.895
40	30	0.118	0.011	-0.008	0.895
40	40	0.114	0.012	-0.007	0.941
40	50	0.106	0.016	-0.015	0.895
50	10	0.095	0.020	-0.015	0.751
50	20	0.123	0.011	-0.015	0.895
50	30	0.161	0.010	-0.013	0.895
50	40	0.134	0.010	-0.004	0.895
50	50	0.132	0.010	-0.017	1.000

Response-window sensitivity of phase prediction

Median phase gains after contextual adjustment were 0.0028, 0.0042, 0.0127, 0.0121, and 0.0152 for 10-, 20-, 30-, 40-, and 50-ms response windows. The corresponding gains after measured-kinematic adjustment were 0.0023, 0.0028, 0.0073, 0.0077, and 0.0084.

Supplemental Table S4. Phase-representation sensitivity

Adjustment	Phase representation	Penalty	Median gain	Positive labels	Wilcoxon P	Sign-test P
Context	Two harmonics	0.01	0.0123	13/20	0.154	0.263
Context	Two harmonics	0.1	0.0137	12/20	0.036	0.503
Context	Two harmonics	1	0.0152	15/20	<0.001	0.041
Context	Two harmonics	10	0.0054	15/20	<0.001	0.041
Context	First harmonic	0.1	0.0094	12/20	0.064	0.503
Context	First harmonic	1	0.0097	15/20	0.002	0.041
Context	First harmonic	10	0.0022	15/20	<0.001	0.041
Context	Separate branches	0.1	0.0081	13/20	0.053	0.263
Context	Separate branches	1	0.0055	16/20	<0.001	0.012
Context	Separate branches	10	0.0028	16/20	<0.001	0.012
Measured kinematics	Two harmonics	0.01	0.0070	13/20	0.189	0.263
Measured kinematics	Two harmonics	0.1	0.0137	12/20	0.064	0.503
Measured kinematics	Two harmonics	1	0.0084	15/20	0.002	0.041
Measured kinematics	Two harmonics	10	0.0050	15/20	<0.001	0.041
Measured kinematics	First harmonic	0.1	0.0014	10/20	0.202	1.000
Measured kinematics	First harmonic	1	0.0040	13/20	0.017	0.263
Measured kinematics	First harmonic	10	0.0018	15/20	<0.001	0.041
Measured kinematics	Separate branches	0.1	-0.0007	10/20	0.648	1.000
Measured kinematics	Separate branches	1	0.0027	14/20	0.027	0.115
Measured kinematics	Separate branches	10	0.0020	16/20	<0.001	0.012

The median gain from the primary two-harmonic model was positive at every penalty strength. Directional consistency and signed-rank support varied across settings, so the manuscript treats the phase contribution as small and dependent on model specification.

Supplemental Table S5. Continuous movement and contact coefficients

The table reports archive-label median standardized coefficients from the 50-ms neighboring-onset set. Q values control the false discovery rate across the eight coefficients. Label-level values are supplied in the numerical source data.

Measure	Archive subject labels	Median beta	Bootstrap 95% CI	q
Whisking amplitude	20	0.079	-0.012 to 0.107	0.145
Pre-touch angular velocity	20	0.039	-0.003 to 0.107	0.432
Pre-touch angular acceleration	20	0.021	-0.002 to 0.049	0.145
Signed peak curvature change	20	0.038	-0.014 to 0.078	0.440
Absolute peak curvature change	20	0.029	-0.041 to 0.127	0.432
Touch duration	20	0.035	-0.042 to 0.094	0.440
Distance to pole	20	-0.007	-0.043 to 0.025	0.801
Base angle at touch	20	-0.067	-0.179 to 0.138	0.927

Contact duration by phase branch

The primary phase set contained 10,113 protraction touches and 2,660 retraction touches. Contact duration was observed for 10,104 protraction touches and 2,642 retraction touches. Median contact duration was 18 ms during protraction and 44 ms during retraction. Among observed durations, 17.1% of protraction touches and 45.5% of retraction touches lasted at least 50 ms. The neighbor-onset rule removes closely timed new contacts and permits an indexed contact to continue through part of the response window.

Direct branch contrast

The direct retraction-minus-protraction difference in raw spikes per touch was -0.043 (20 archive subject labels; 95% CI, -0.141 to 0.026; $P = 0.202$). The corresponding difference was -0.117 standardized units after contextual adjustment (95% CI, -0.227 to 0.040; $P = 0.143$) and -0.030 after measured-kinematic adjustment (95% CI, -0.169 to 0.047; $P = 0.245$). These estimates compare weighted branch means within each recording file, followed by one median per archive subject label.

Post-onset contact-description sensitivity

When signed and absolute curvature change and contact duration were added to the predictive adjustment, the median gain from the two-harmonic phase terms was 0.0038 (20 archive subject labels; 12 positive; 95% CI, -0.0023 to 0.0151; $P = 0.070$). The post-onset variables overlap the 50-ms response interval, so this result is reported as a descriptive sensitivity.

Recording-location sensitivity

The primary measured-kinematic model yielded a median phase gain of 0.0210 in 25 C2 files from 12 archive subject labels (95% CI, 0.0084 to 0.0348) and 0.0017 in 10 surrounding-barrel files from 9 labels (95% CI, -0.0035 to 0.0075). One label contributed files to both groups, so the estimates are descriptive subgroup summaries.